



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

February 13, 2020

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours 30 minutes

Prof. Marcello Restelli

Maximum Marks: 34

- The following exam is composed of **10 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam**. If you finish earlier than 45 min before the end of the exam you will get 3 points, if you finish earlier than 30 min you will get 2 points and if you finish earlier than 15 min you will get 1 point (the points cannot be accumulated).
- The box on Page 12 can only be used to complete the Exercises 9 and/or 10.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Ex. 9	Ex. 10	Time	Tot.
/ 5	/ 5	/ 5	/ 2	/ 2	/ 2	/ 2	/ 2	/ 3	/ 3	/ 3	/ 34

Exercise 1 (5 marks)

Describe and compare **logistic regression** with the **perceptron algorithm**.

Exercise 2 (5 marks)

Describe what is the **VC-dimension** of a hypothesis space and how it can be used for avoiding over-fitting.

Exercise 3 (5 marks)

Describe two algorithms to perform **model-free policy evaluation**.

Exercise 4 (2 marks)

```
1 load carsmall;
2 X = zscore([Weight Displacement]); t = zscore(Acceleration);
3 w_1 = -2:0.1:3; w_2 = -3:0.1:1;
4 for ii = 1:length(w_1)
5     for jj = 1:length(w_2)
6         loss(ii,jj) = sum((X * [w_1(ii) w_2(jj)]' - t).^2);
7         x(ii,jj) = w_1(ii);
8         y(ii,jj) = w_2(jj);
9     end
10 end
11 [row, col] = find(loss == min(loss(:)));
12 contour(x, y, loss);
13 hold on;
14 plot(w_1(row), w_2(col), 'rx');
```

1. Describe the operations performed by the Matlab code snippet above.
2. Which kind of problem are we solving? What is the technique we are using?
3. Do you think that there exists a better way to perform the procedure in Lines 4-10?

1.
 - Line 1 – 3 Loads the carsmall dataset, normalizes the data and generate two parameters vectors
 - Line 4 – 10 For each combination of the parameters in the two vectors, evaluates the loss of a linear model
 - Line 11 Minimizes the loss.
 - Line 12 – 14 Plot a contour of the loss function in the parameter space.
2. It is a regression problem, and we are solving it with the Least Square method.
3. A more efficient technique would be to use the closed form solution for the problem.

Exercise 5 (2 marks)

We selected the features of a linear regression model by running a forward feature selection procedure which minimizes the validation error. Tell if the following elements increase or decrease in the considered model w.r.t. the one in the model that uses all the features, when both models are trained on the same data:

1. The sum of the squared residuals on the same training set;
2. The variance;
3. The squared bias;
4. The sum of the squared residuals on a test set.

1. INCREASES: since we are removing some features, the optimal solution is for sure worse or equal than the original one.
2. DECREASES: we are selecting a simpler model which has a less variance.
3. INCREASES: since the behaviour of the bias is opposite to the one of variance.
4. DO NOT KNOW: it is likely to decrease as in the validation set used to select the model, but we cannot predict for sure what its behaviour will be.

Exercise 6 (2 marks)

Tell if the following statements about solving Markov Decision Processes (MDPs) are true or false.

1. There exists a unique optimal policy in an MDP;
2. In a finite state MDP there always exists a Markovian, stationary and deterministic optimal policy;
3. The results of coupling a specific policy and an MDP is a Markov process;
4. The action-value function does not contain all the information to execute the optimal policy on a given MDP.

Motivate your answers.

1. FALSE: there might exist multiple policies which provide the same cumulative reward, for instance if two different actions in a state provide the same instantaneous reward and have the same transition probabilities.
2. TRUE: we are assured (theorem) that at least one of these optimal policies exists.
3. TRUE: we are removing the choice of the action from an MDP, and therefore we are transforming it into a Markov process.
4. FALSE: selecting the action which maximizes the action-value function in each state provides the optimal policy.

Exercise 7 (2 marks)

Tell if the following statements about Q-learning and SARSA are true or false.

1. SARSA is an online Reinforcement Learning method;
2. Q-learning is an on-policy method;
3. The update for the Q-learning can be equivalent to the SARSA one in specific cases;
4. If the exploration policy has the GLIE property, the SARSA algorithm is guaranteed to converge to the optimal value function.

Motivate your answers.

1. TRUE: the SARSA algorithm is able to update the estimates about the cumulative reward in an MDP as soon as a new step in an episode is provided.
2. FALSE: the Q-learning algorithm is able to find the optimal policy even if trained on data coming from a different policy.
3. TRUE: if the greedy policy corresponds to the one prescribed by Q-learning then both would perform the update of the estimates (either value function or state-action function) in the same way.
4. FALSE: we also need to have an assumption on the step size (theorem seen during lecture).

Exercise 8 (2 marks)

Categorize the following Machine Learning problems:

1. Identifying the best restaurant as fast as possible by repeatedly trying them;
2. Maximize the revenue of a stock portfolio over time;
3. Recognize a person from their picture;
4. Identify the communities of users based on their internet purchase habits.

For each one of them suggest **a set of features** that might be useful to solve the problem and **a method** to solve it.

1. Multi-armed bandit, since we are faced with a choice among a finite number of options and we can interact with the environment by selecting them over a number of turns. Features one might use (for instance for a contextual MAB) are the kind of restaurant, the location. A techniques to solve these problem has not been presented during the course, but for standard MAB one might use UCB1 or TS.
2. Reinforcement Learning problem, since the action we perform in the past are influencing the ones we will perform in the future. We can use as features for the modeling of the state some financial index providing insights about the market behaviour. We can use Q-learning to determine the optimal policy.
3. Classification problem, since we want to decide the label (person) basing on some features contained in the picture. In this case one might use the entire picture as features vector or extract some characteristics of the face (either manually or automatically). Using NN for classification is a good idea.
4. Clustering, since we want to divide the set of user based on their habits. The features might be about the frequency and average amount of the transaction they make as well as the kind of purchase they perform. A method to perform is K-means (not discussed during lectures).

Exercise 9 (3 marks)

Consider a hard margin SVM for classification. After the training phase, you get the following support vectors $\mathbf{x}_1 = [4 \ 1]^\top$, $\mathbf{x}_2 = [6 \ 3]^\top$, $\mathbf{x}_3 = [3 \ 4]^\top$, and $\mathbf{x}_4 = [1 \ 6]^\top$, with the corresponding labels $t_1 = t_2 = -1$, and $t_3 = t_4 = 1$, and weights $\alpha_1 = 1/8$, $\alpha_2 = 1/8$, $\alpha_3 = 1/4$, $\alpha_4 = 0$, and a bias $b = -1/2$. Answer the following questions and motivate:

1. Plot and provide the analytic formula of the boundary and the margins determined by the SVM in the input space (hint: start by plotting the points in the training set);
2. If we had as additional point $\mathbf{x}_5 = [2 \ 2]^\top$, $t_5 = 1$ in the training set, would you consider retraining the SVM?
3. Do you think that this problem would benefit from the use of a non-linear kernel in the SVM formulation?

Exercise 10 (3 marks)

Given the following dataset:

$$\mathbf{x}_1 = [2 \ 3 \ 4], \ y_1 = 1$$

$$\mathbf{x}_3 = [1 \ 2 \ 5], \ y_3 = 1$$

$$\mathbf{x}_5 = [1 \ 3 \ 1], \ y_5 = 0$$

$$\mathbf{x}_7 = [3 \ 1 \ 4], \ y_7 = 1$$

$$\mathbf{x}_9 = [1 \ 3 \ 3], \ y_9 = 0$$

$$\mathbf{x}_2 = [0 \ 1 \ 2], \ y_2 = 0$$

$$\mathbf{x}_4 = [1 \ 4 \ 3], \ y_4 = 0$$

$$\mathbf{x}_6 = [1 \ 2 \ 2], \ y_6 = 0$$

$$\mathbf{x}_8 = [4 \ 2 \ 5], \ y_8 = 1$$

$$\mathbf{x}_{10} = [1 \ 2 \ 4], \ y_{10} = 1$$

1. Classify the point $\mathbf{x}_{11} = [0 \ 1 \ 2]$ according to a KNN classifier trained on the given dataset with $K = 3$ and using the euclidean distance;
2. What happens if we use $K = 10$ instead? Do you think it is a good idea to choose such a parameter (hint: two pros if it is a good idea or two cons if it is not);
3. Suggest a technique to set the parameter K .

1. Computing the euclidean distance from each point to the given one, we have that the point $\mathbf{x}_2$ is the nearest one (it coincides with $\mathbf{x}_{11}$) and the second nearest is $\mathbf{x}_6$. Since both are belonging to the negative class, even without checking the third nearest point we can say that the KNN with $K = 3$ would predict the point $\mathbf{x}_{11}$ to be in the negative class, $\hat{y}_{11} = 0$.
2. It is not a good idea since in this case we would have exactly the same results for each new datapoint (a tie since the dataset has the same amount of positive and negative samples). Moreover, in general, it is a good practice to select an odd value for the K parameter to avoid ties.
3. We can use cross-validation (k-fold) to select the K having the lowest expected error.

